DSC 152:
Applied Statistical Data Analysis and Inference

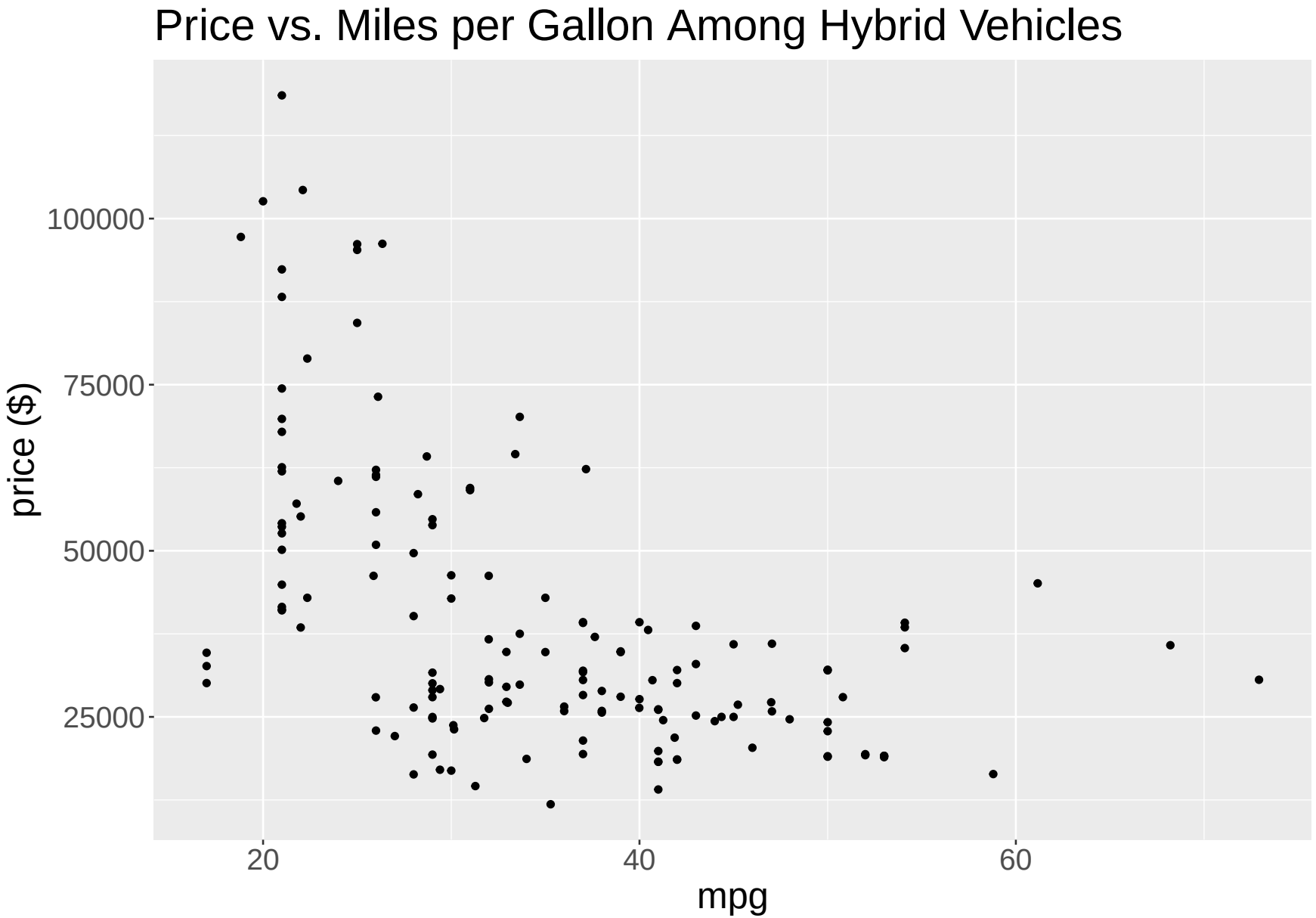
Lecture #14
Transformations

Thursday, November 12
Fall Quarter 2026
Armin Schwartzman

Recall: Conditions for Linear Regression

What if the relationship looks like this?

A dataset from DSC 10



What if the relationship looks like this?

- The relationship is clearly not linear!
- Are we screwed?

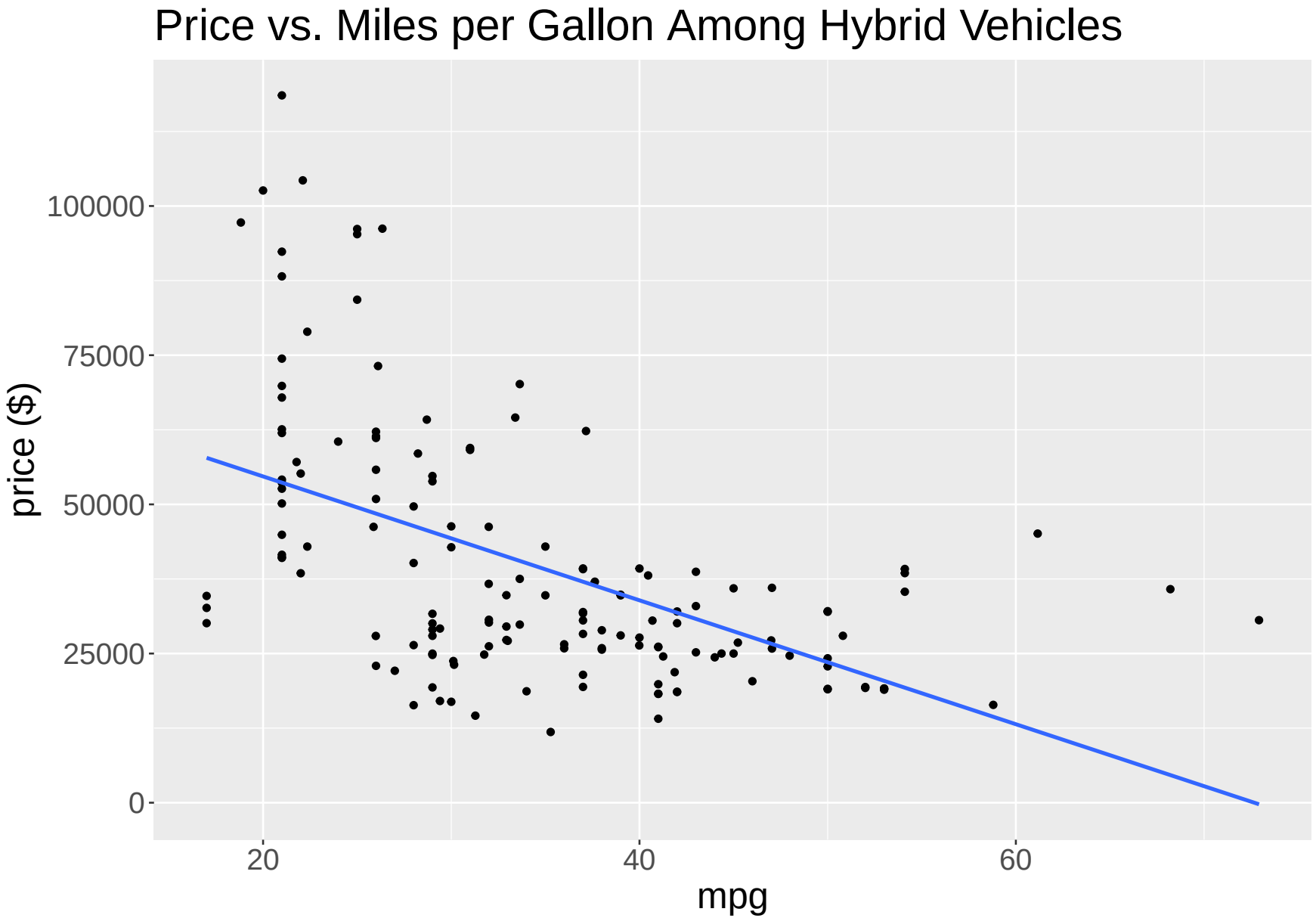
We can still run a linear model...

```r
model1 <- lm(price ~ mpg, data=hybrid)
summary(model1)
```

```
##
## Call:
## lm(formula = price ~ mpg, data = hybrid)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -30033 -12199  -2221   8315  64899
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  75448.2     4907.5  15.374  < 2e-16 ***
## mpg          -1038.3      134.5  -7.717 1.51e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 18200 on 151 degrees of freedom
## Multiple R-squared:  0.2828, Adjusted R-squared:  0.2781
## F-statistic: 59.55 on 1 and 151 DF,  p-value: 1.507e-12
```

But clearly this line is ridiculous

Anyone want a free car that gets 75 MPG??



But clearly this line is ridiculous

What's wrong with fitting a linear model to non-linear data?

- The line will be a terrible fit at certain (if not all) values of the predictor variable (as seen on the previous slide)
- Validity of inference is in question! (as we saw in Lab 4)

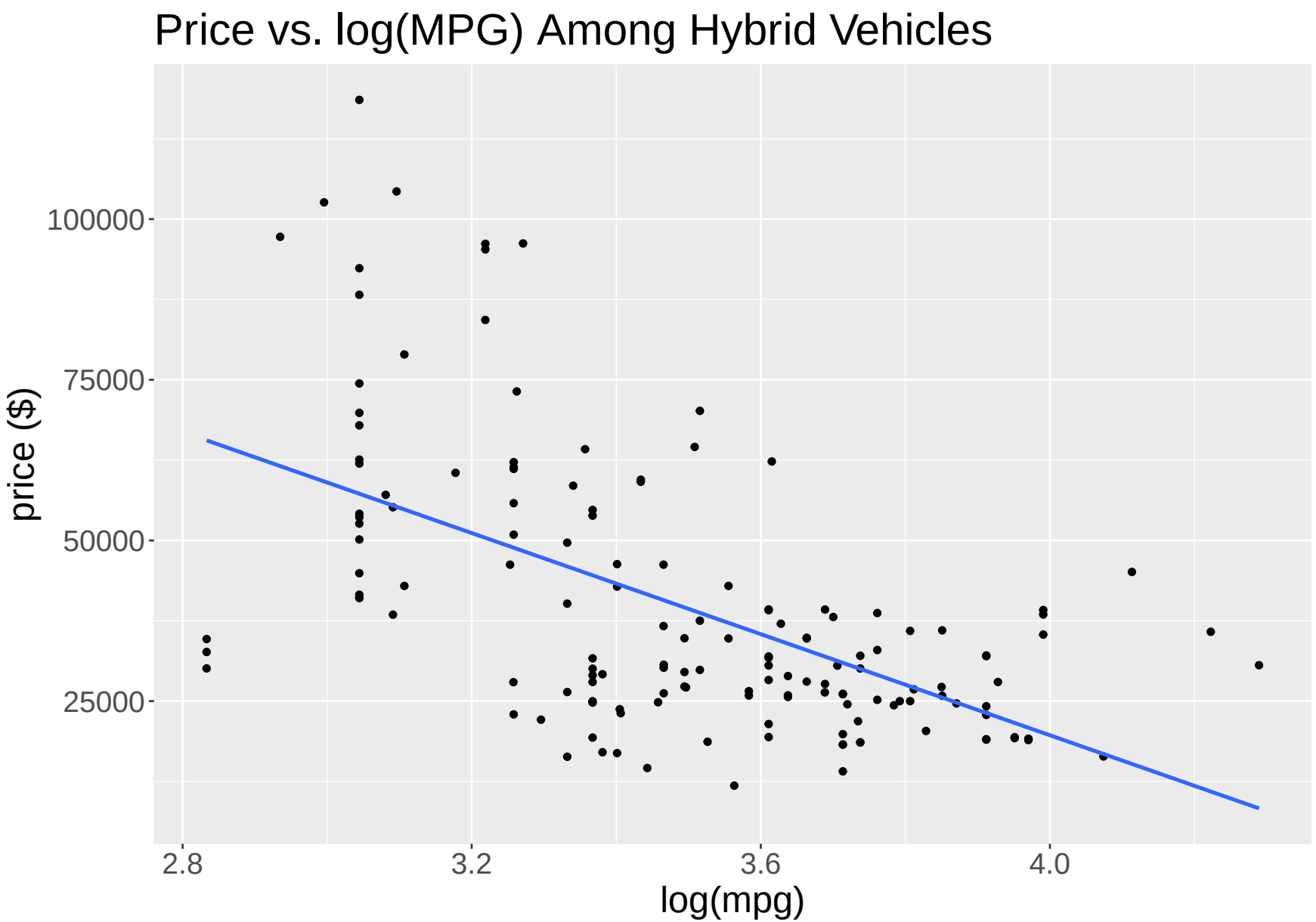
What do transformations do for us?

If we can find a function $g(x)$ such that y has a linear relationship with $g(x)$, then we can proceed with fitting:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 \cdot g(x)$$

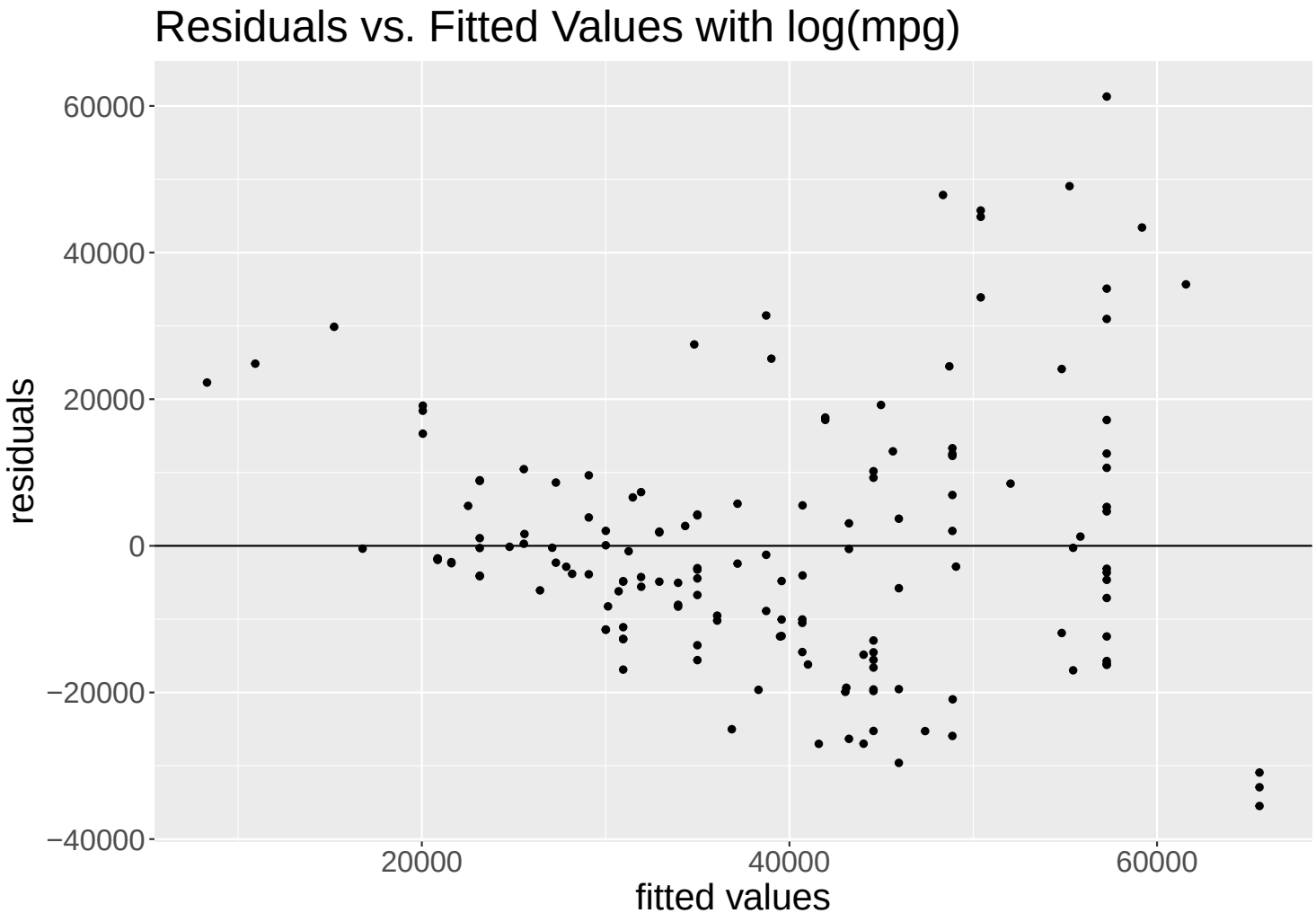
This is then still LINEAR regression (it is linear with respect to some function of x)! And all of the benefits of linear regression then apply again.

(natural) Log transformation of MPG



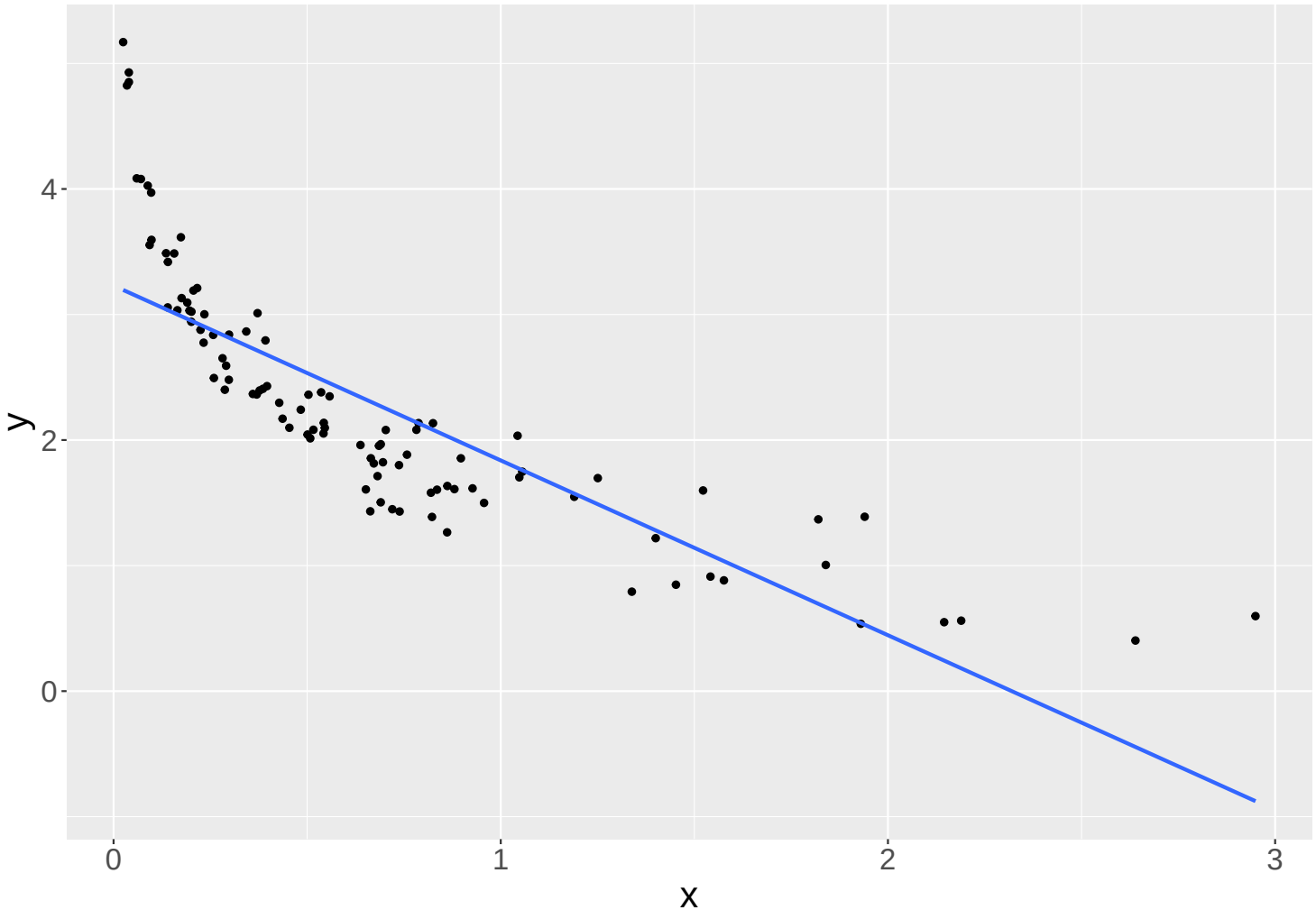
Log transformation of MPG

This didn't help much here. Here is the residual plot where it is pretty obvious that this is still not good:



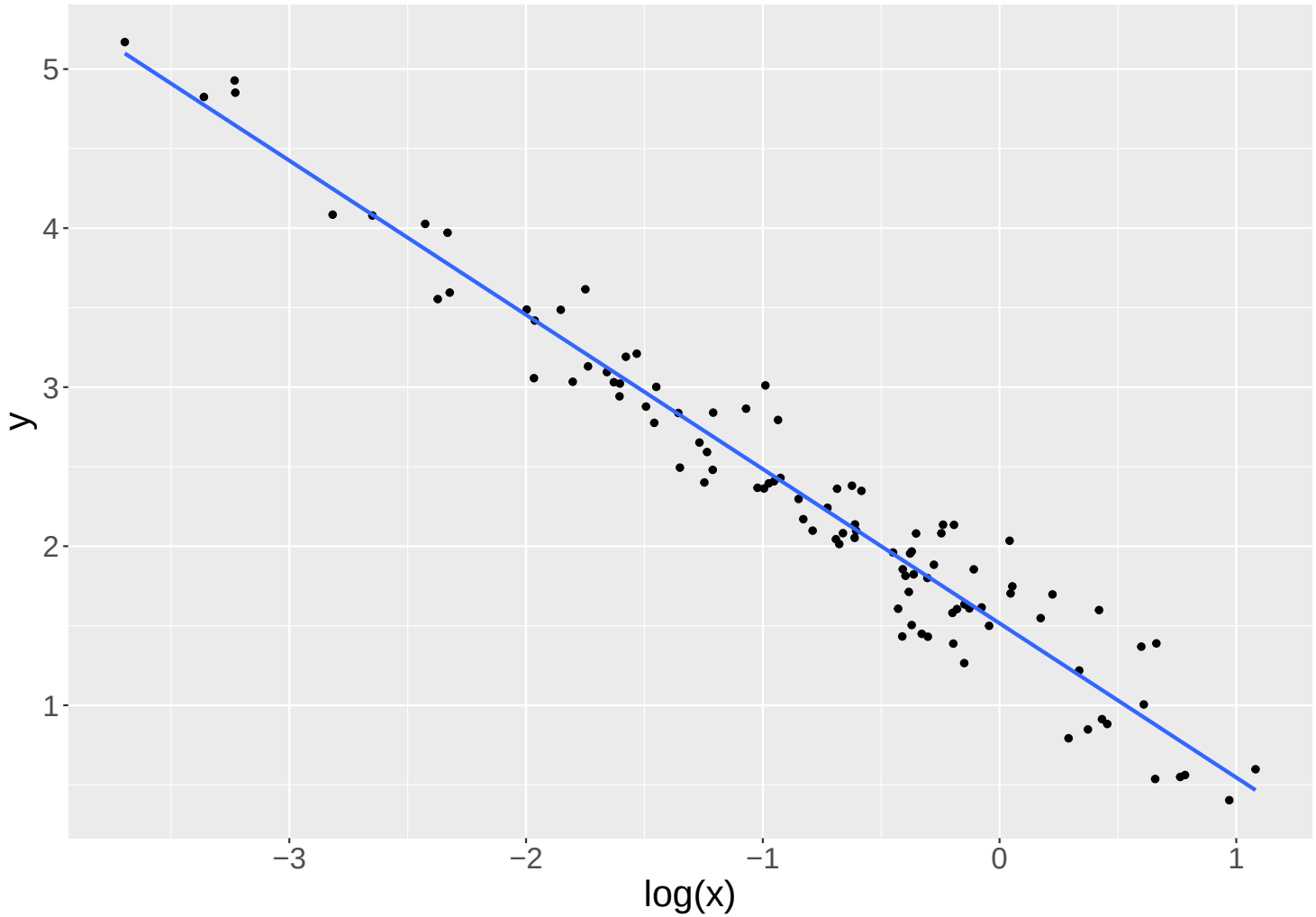
Log transformation of x

Now here is a (simulated) example where log-transforming the x-variable works really well:



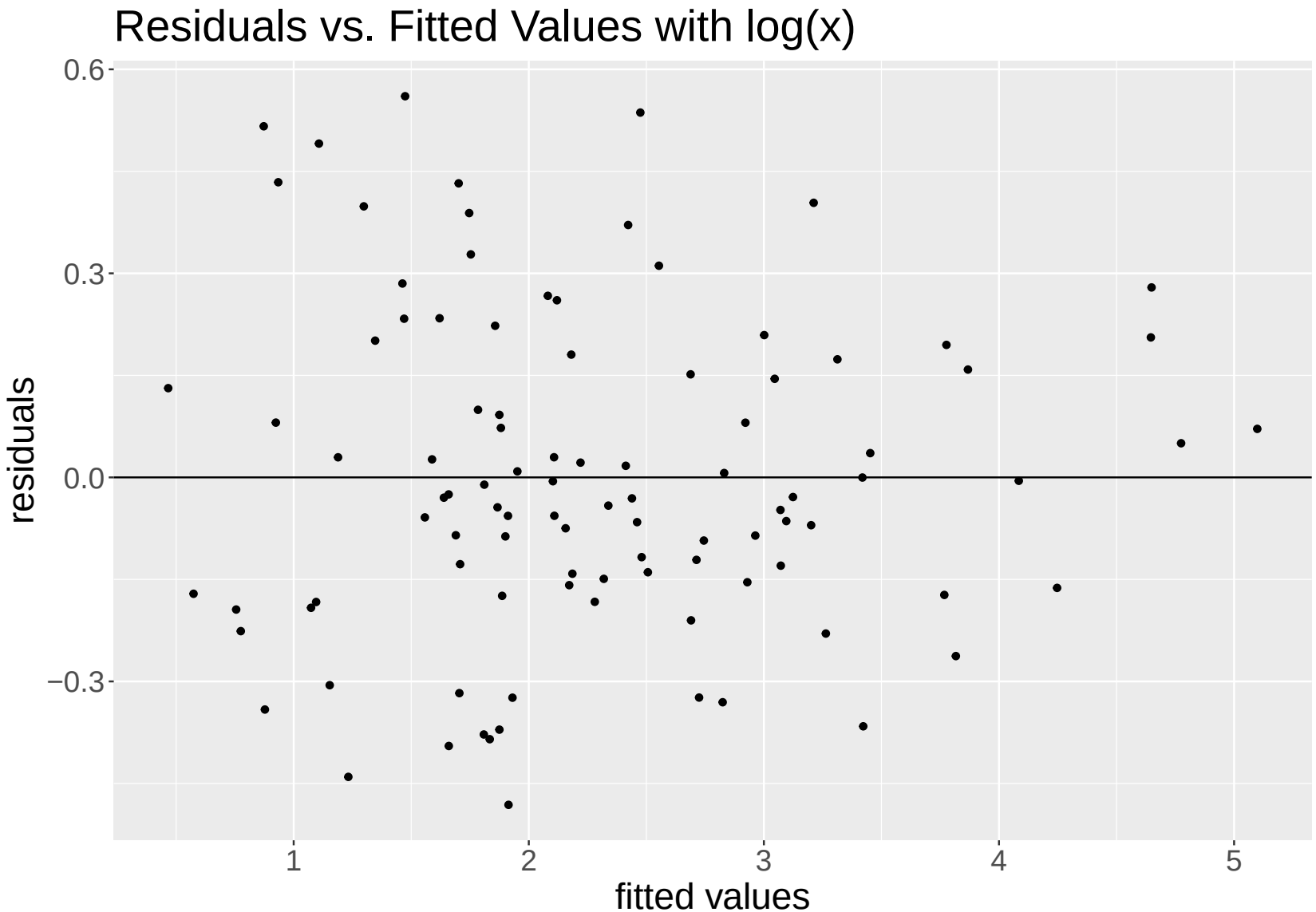
Log transformation of x

Now here is a (simulated) example where log-transforming the x-variable works really well:



Log transformation of x

And the residual plot (things look good here):



Log transformation of x

Important note: this is still LINEAR regression!

This is true even though we are modeling a curved relationship between y and x. How??

It is because we are modeling a linear relationship between y and $\log(x)$.

Also note that it is a convention among statisticians to use "log" to refer to the natural log or "ln." We will refer to the base-10 log, if ever used, as "$\log_{10}$."

Two Questions:

- What is the interpretation of the slope coefficient now?
- What are the benefits of doing the transformation, in terms of statistical inference?

Log transformation of x

Coefficient interpretation

```
##
## Call:
## lm(formula = y ~ log(x), data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.48146 -0.15947 -0.02937  0.16230  0.56042
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.51529    0.03061   49.51   <2e-16 ***
## log(x)      -0.96978    0.02397  -40.46   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2392 on 98 degrees of freedom
## Multiple R-squared:  0.9435, Adjusted R-squared:  0.9429
## F-statistic:  1637 on 1 and 98 DF,  p-value: < 2.2e-16
```

Log transformation of x

Coefficient interpretation

The value of -0.9698 is...

Log transformation of x

Statistical Power

Question: If we correctly fit a linear model with $\log(x)$ instead of just x, do we benefit in terms of statistical power?

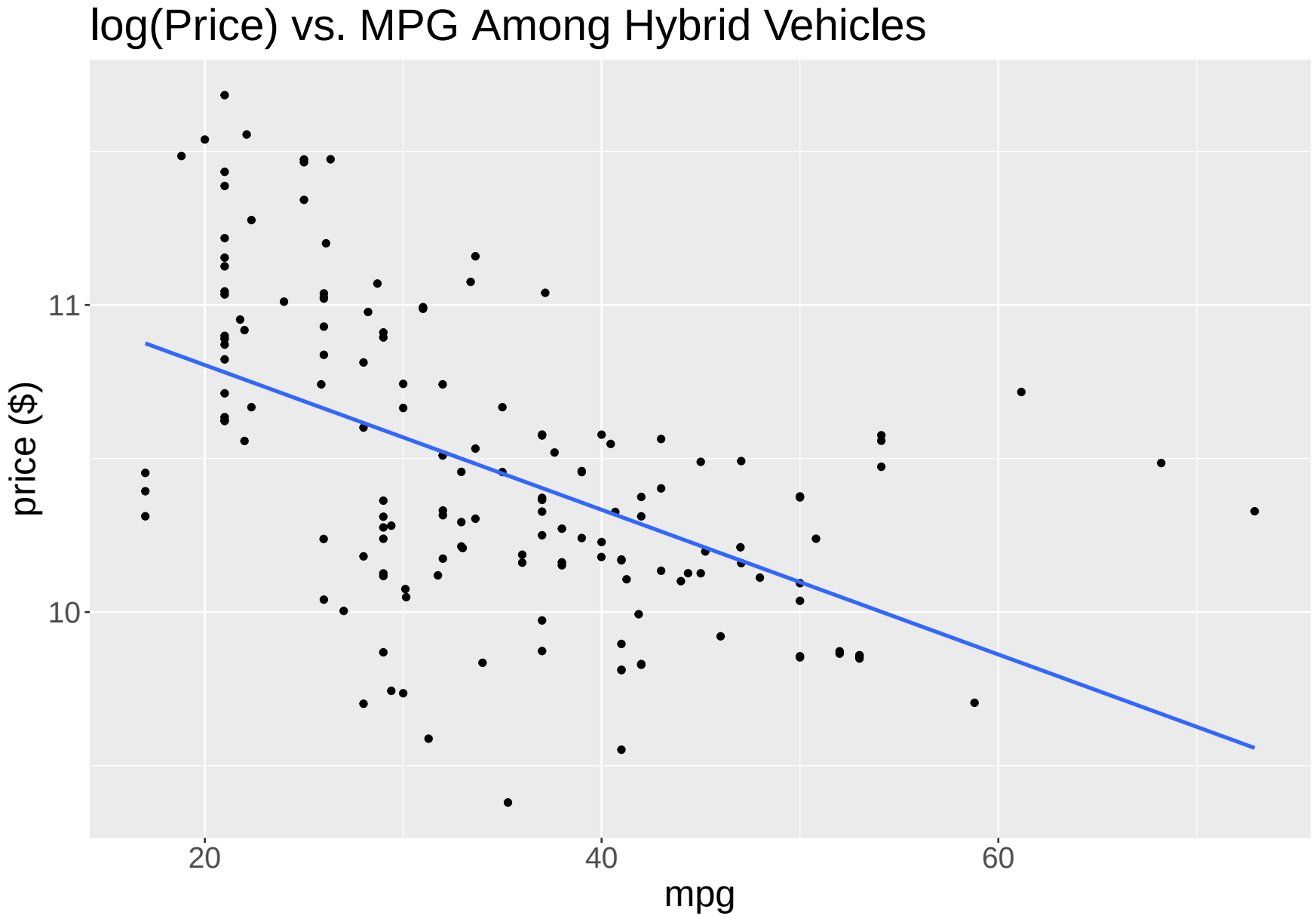
Your Turn (#1?): Simulation of power

- Simulate 100 values of $x \sim \text{Unif}(0, 3)$
- Generate values of $y = \log(x) + \epsilon$ where $\epsilon_i \sim N(\mu = 0, \sigma = 3)$
- Run these two linear models:
  - $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 \cdot x$
  - $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 \cdot \log(x)$
- Check whether $H_0 : \beta_1 = 0$ is rejected in each case
- Do this repeatedly to get estimates of power for each model

Comment briefly on what you observe.

Back to the Hybrid Vehicle Dataset

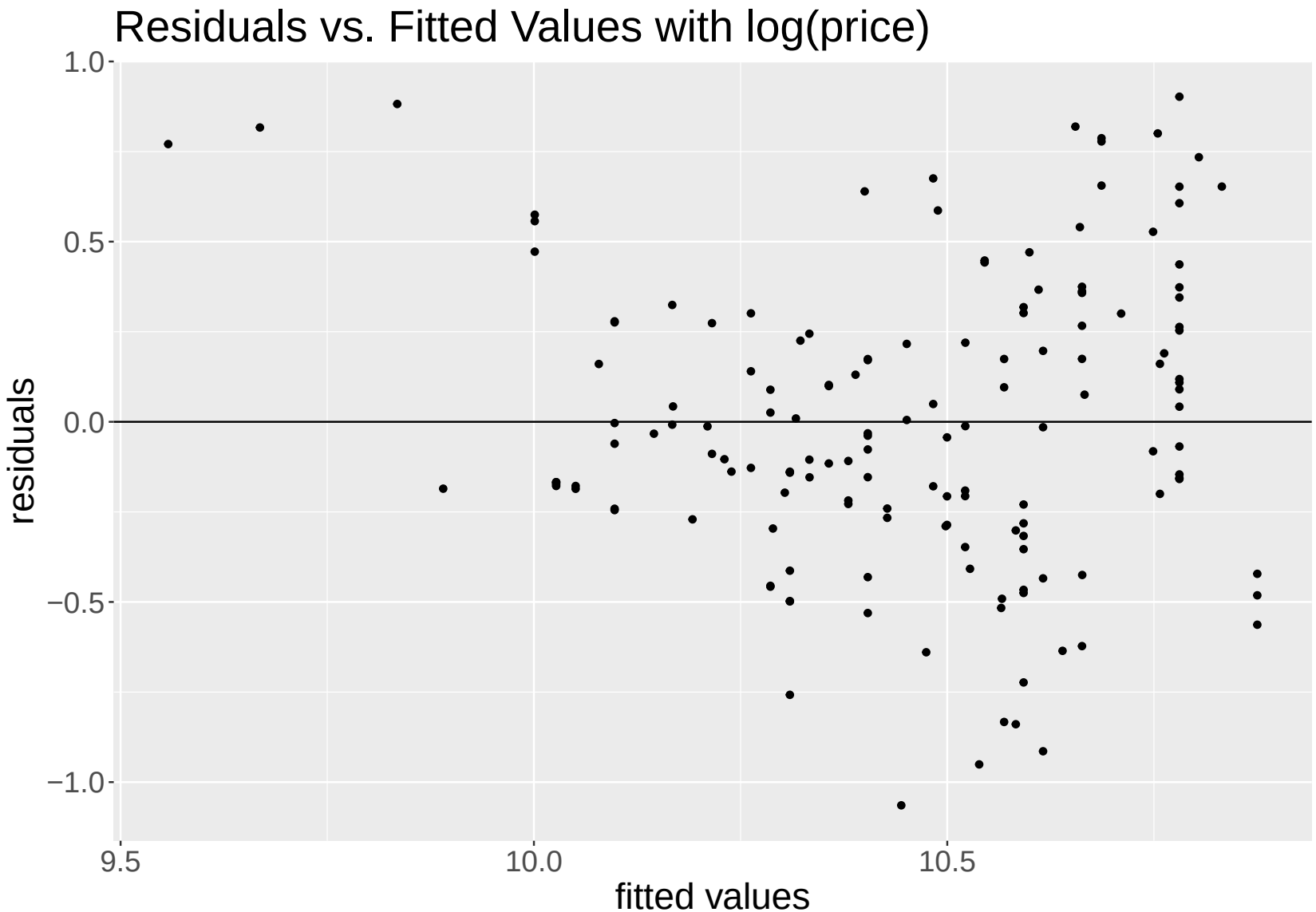
Log transformation of price



Back to the Hybrid Vehicle Dataset

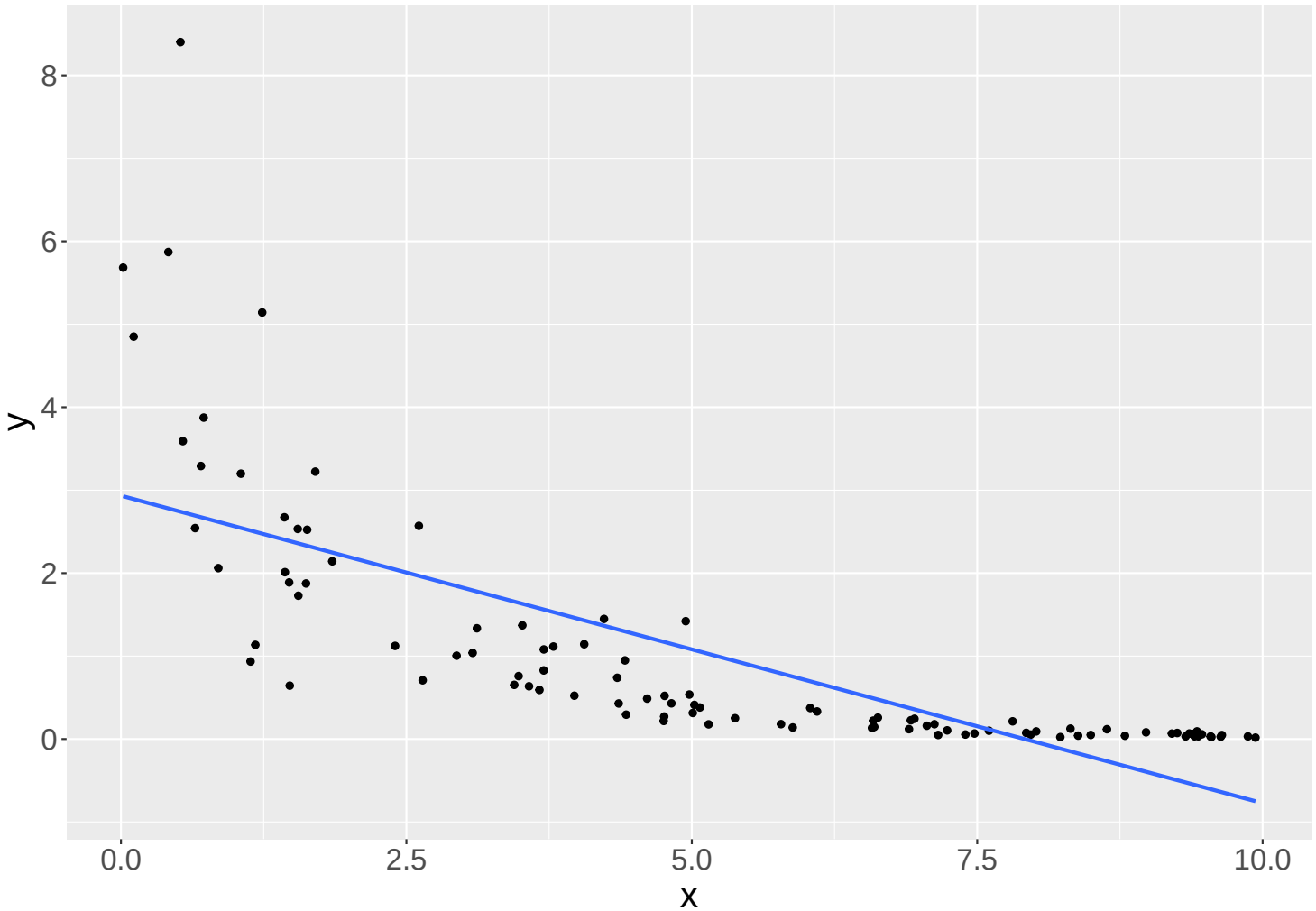
Log transformation of price

This didn't help much either:



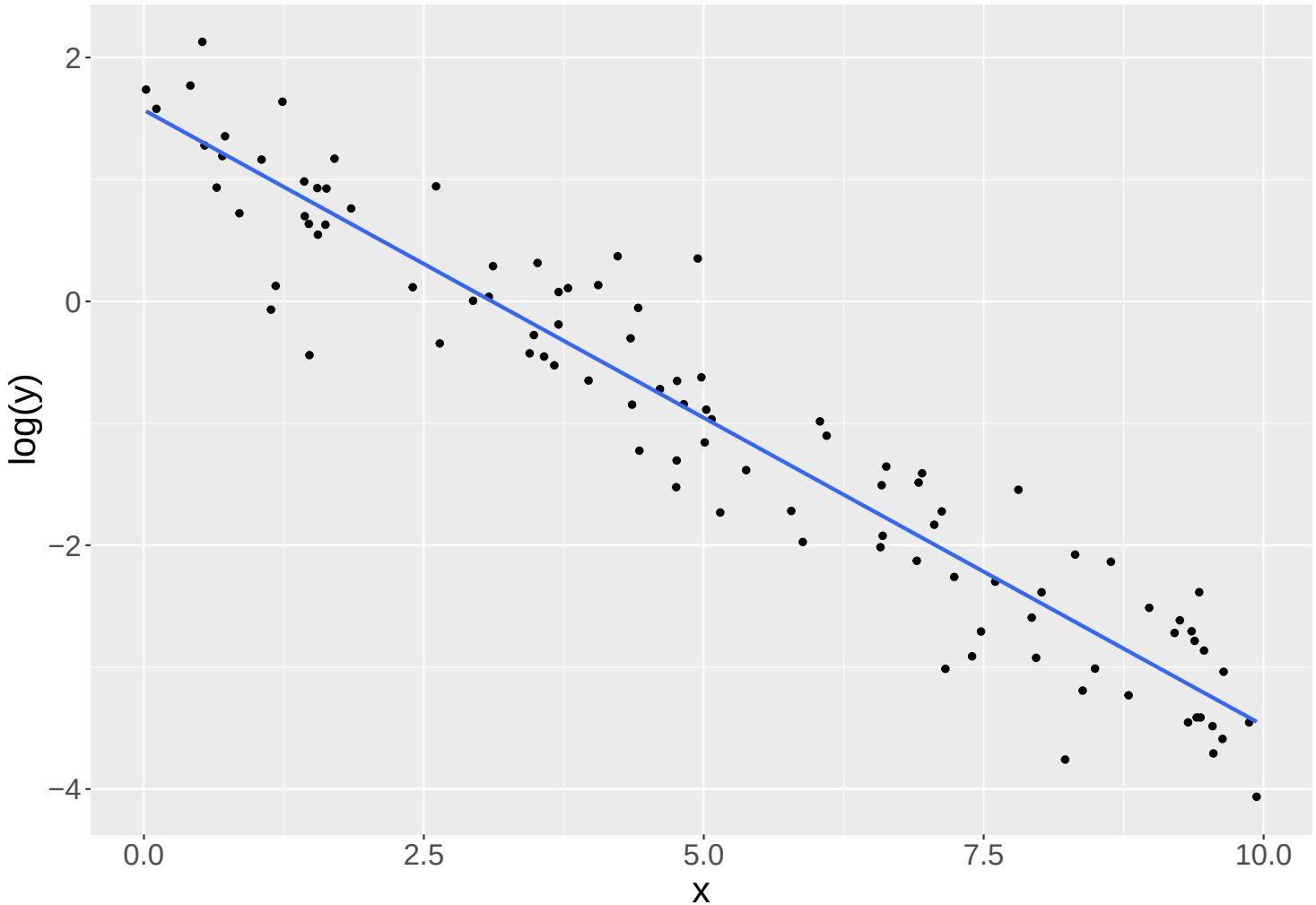
Log transformation of y

Now here is a (simulated) example where log-transforming the y-variable works really well:



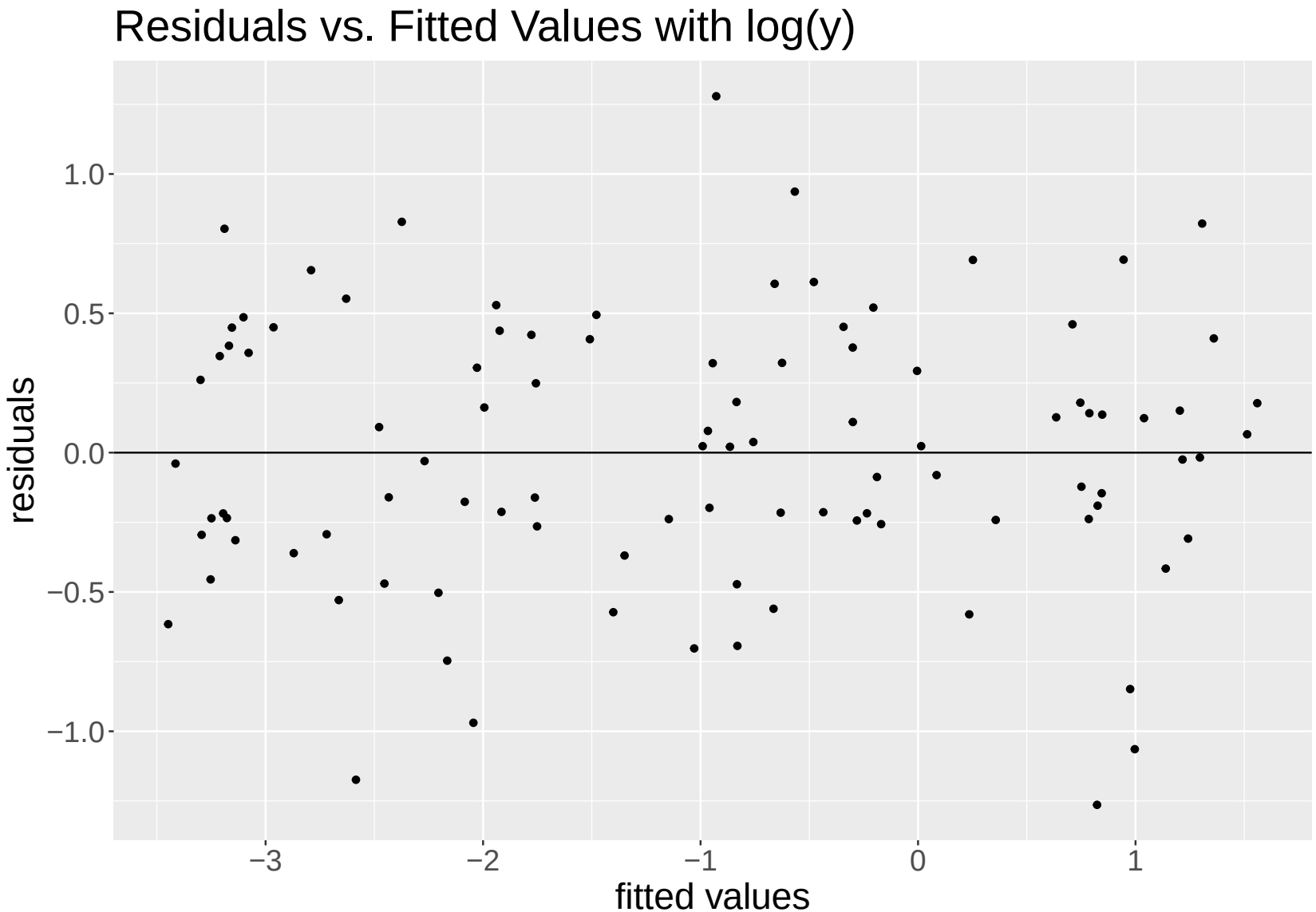
Log transformation of y

Now here is a (simulated) example where log-transforming the y-variable works really well:



Log transformation of y

And the residual plot (things look good here):



Log transformation of y

Coefficient interpretation

```
##
## Call:
## lm(formula = log(y) ~ x, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.26401 -0.25859  0.00181  0.36290  1.27910
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.56970    0.09521   16.49   <2e-16 ***
## x           -0.50490    0.01597  -31.61   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4779 on 98 degrees of freedom
## Multiple R-squared:  0.9107, Adjusted R-squared:  0.9098
## F-statistic: 999.4 on 1 and 98 DF,  p-value: < 2.2e-16
```

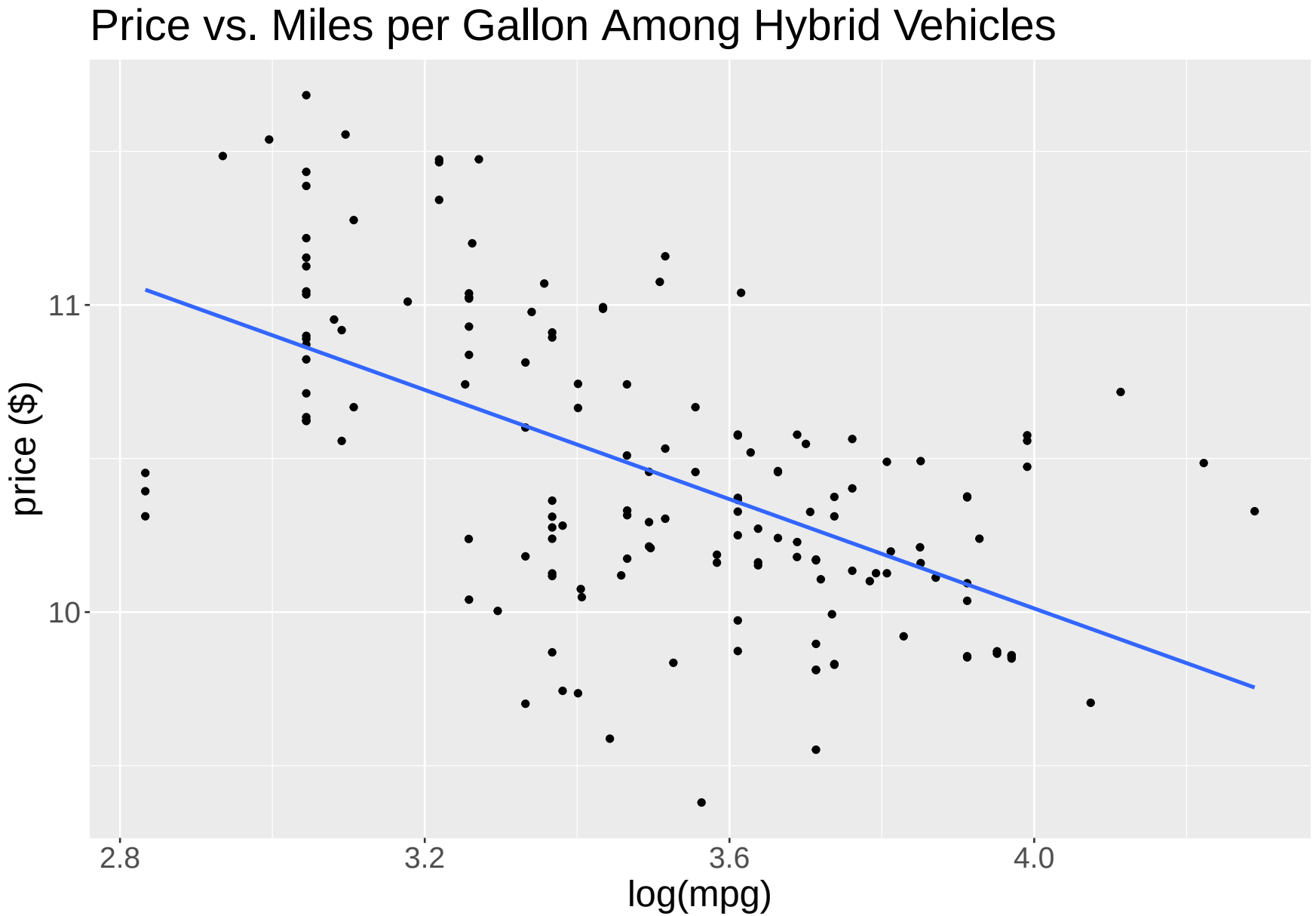
Log transformation of y

Coefficient interpretation

The value of -0.5049 is. . .

Back to the Hybrid Vehicle Dataset Again

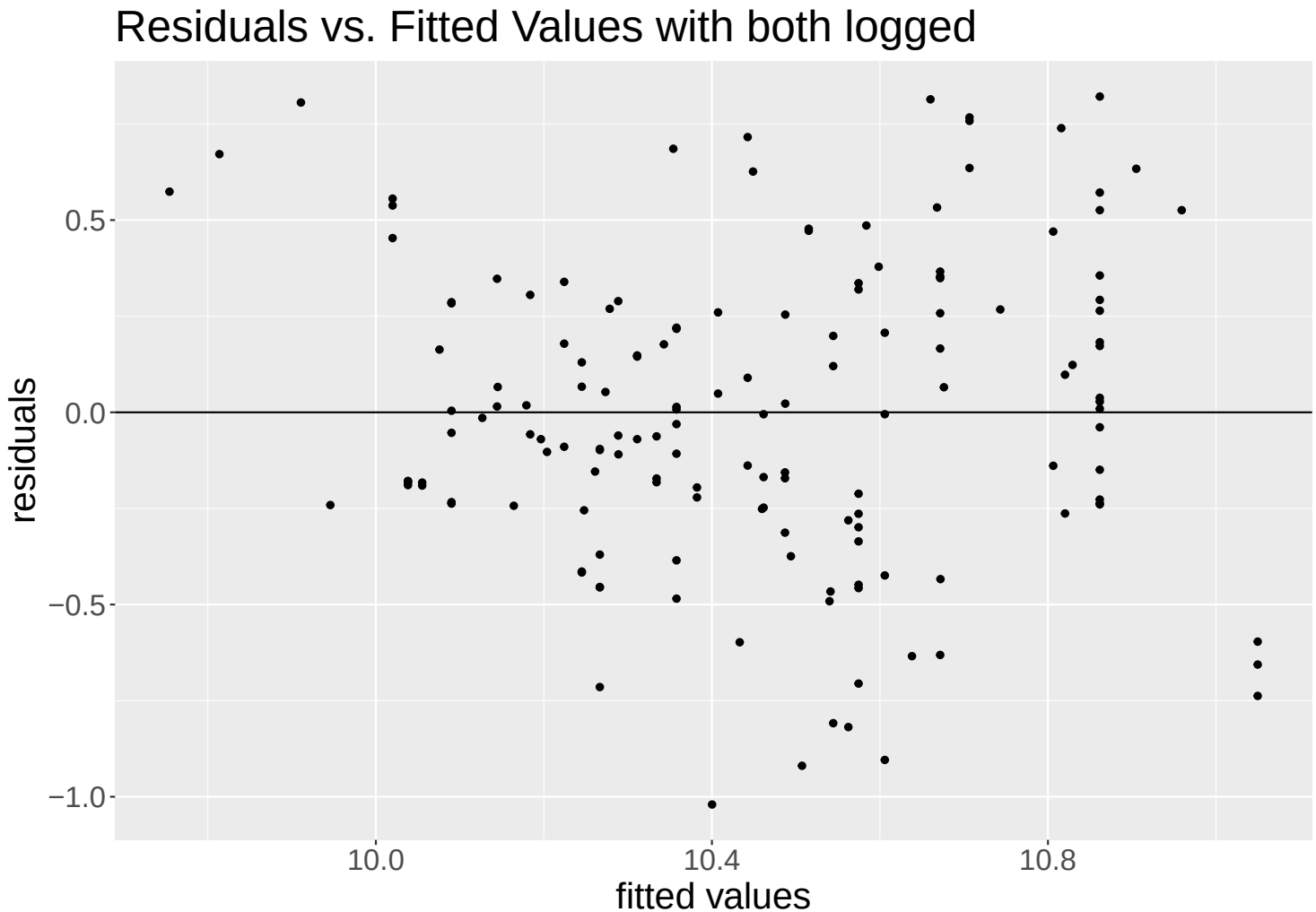
Log transformation of both



Back to the Hybrid Vehicle Dataset Again

Log transformation of both

This is actually somewhat ok!



Back to the Hybrid Vehicle Dataset Again

Log transformation of both

```
##
## Call:
## lm(formula = log(price) ~ log(mpg), data = hybrid)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.02017 -0.23954 -0.00494  0.26923  0.82136
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  13.5698     0.3609  37.596  < 2e-16 ***
## log(mpg)     -0.8895     0.1027  -8.662 6.58e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3978 on 151 degrees of freedom
## Multiple R-squared:  0.332,  Adjusted R-squared:  0.3275
## F-statistic: 75.03 on 1 and 151 DF,  p-value: 6.582e-15
```

Back to the Hybrid Vehicle Dataset Again

Log transformation of both

Coefficient interpretation

The value of -0.8895 is. . .

Centering and Scaling

Recall the FEV dataset and the interaction model from Lecture #12:

```
##
## Call:
## lm(formula = fev ~ smoke * age, data = FEV)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.76645 -0.34947 -0.03364  0.33679  2.05990
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.253396   0.082651   3.066  0.00226 **
## smoke        1.943571   0.414285   4.691 3.31e-06 ***
## age          0.242558   0.008332  29.113  < 2e-16 ***
## smoke:age   -0.162703   0.030738  -5.293 1.65e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5537 on 650 degrees of freedom
## Multiple R-squared:  0.5941, Adjusted R-squared:  0.5922
## F-statistic: 317.1 on 3 and 650 DF,  p-value: < 2.2e-16
```

Centering and Scaling

From this output, in one sense it appears that smoking status has a much larger impact on FEV than a child's age does (since its coefficient value is much larger than the coefficient for age).

But, is this really the case?

Centering and Scaling

```
#{r, echo=FALSE, message=FALSE, size="footnotesize", warning
```

```
##
## Call:
## lm(formula = fev ~ smoke * z.age, data = FEV)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.76645 -0.34947 -0.03364  0.33679  2.05990
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.66229    0.02305 115.490  < 2e-16 ***
## smoke        0.32774    0.12861   2.548   0.0111 *
## z.age        0.71650    0.02461  29.113  < 2e-16 ***
## smoke:z.age -0.48061    0.09080  -5.293 1.65e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5537 on 650 degrees of freedom
## Multiple R-squared:  0.5941, Adjusted R-squared:  0.5922
## F-statistic: 317.1 on 3 and 650 DF,  p-value: < 2.2e-16
```

Centering and Scaling

When we center and scale the covariates, a few things happen:

1) Coefficients become more directly comparable to each other
2) The intercept term is now the estimated value of the outcome variable at the average value of the predictor variables (as opposed to when the predictor variables equal 0)
3) The individual main effect p-values are for that covariate at the average value of the other covariate (as opposed to when the other covariate is equal to 0)